

REVIEW EX

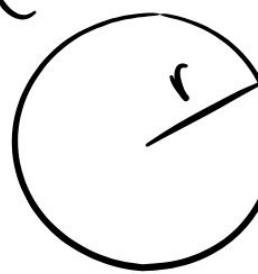
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4

$$\frac{b}{c} = 1$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$y - y_1 = m(x - x_1)$$



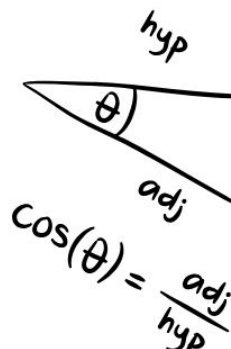
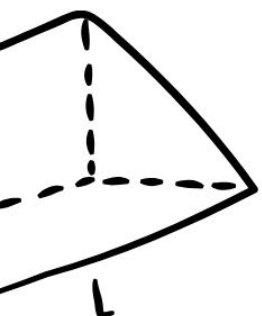
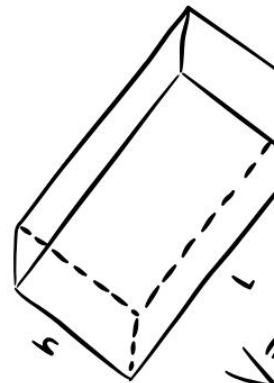
$$C = 2\pi r$$

$$S = \frac{d}{t}$$

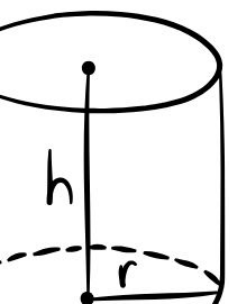
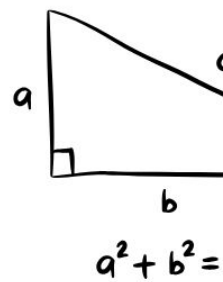
$$\frac{+}{-} = -$$

$$\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\frac{V_f - V_i}{+}$$



bhl



$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\sqrt{b^2 - 4ac}$$



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REVIEW Ex: 4

(82)

The 5th term of an A.P is 60 and 8th term is 90 - Find 12th term.

Sol

Given that:

$$\Rightarrow a_5 = 60 \text{ --- (1)}$$

$$\Rightarrow a_8 = 90 \text{ --- (2)}$$

$$\Rightarrow a_{12} = ?$$

Take (1)

$$\Rightarrow a_5 = 60$$

$$\Rightarrow a + (5-1)d = 60$$

$$\Rightarrow a + 4d = 60 \text{ --- (3)}$$

Take (2)

$$\Rightarrow \therefore a + (n-1)d = a_n$$

$$\Rightarrow a + (8-1)d = 90$$

$$\Rightarrow a + 7d = 90 \text{ --- (4)}$$

Subtract (4) from (3)

$$a + 4d = 60$$

$$a + 7d = 90$$

$$-3d = -30$$

$$\boxed{d = 10}$$

Put d in (3)

$$\therefore (3) \Rightarrow a + 4(10) = 60$$

$$\Rightarrow a = 60 - 40$$

$$\boxed{\Rightarrow a = 20}$$

For a_{12} :

$$\therefore a_n = a + (n-1)d$$

here,

$$a = 20, d = 10, n = 12$$

$$\therefore a_{12} = 20 + (12-1)10$$

$$\Rightarrow a_{12} = 20 + (11)(10)$$

$$\Rightarrow a_{12} = 20 + 110$$

$$\Rightarrow a_{12} = 130$$

It is the required 12th term.

Q.E.D.

- ③ There are n A.Ms b/w 8 and 32 such that $A_3 : A_7 = 3 : 5$ - Find value of n .

Sol

$$8, A_1, A_2, A_3, \dots, A_n, 32$$

here,

$$a = 8, b = 32$$

$$\Rightarrow A_3 : A_7 = 3 : 5$$

$$\text{or } \frac{A_3}{A_7} = \frac{3}{5} \quad \text{--- (1)}$$

As we know that:

$$A_n = a + nd$$

For d :

$$\Rightarrow d = \frac{b-a}{n+1}$$

$$\Rightarrow d = \frac{32-8}{n+1}$$

$$\Rightarrow d = \frac{24}{n+1}$$

Now,

$$\Rightarrow A_3 = a + 3d$$

$$\Rightarrow A_3 = 8 + 3 \left(\frac{24}{n+1} \right)$$

$$\Rightarrow A_3 = 8 + \frac{72}{n+1}$$

$$\Rightarrow A_3 = \frac{8n+8+72}{n+1}$$

$$\Rightarrow A_3 = \frac{8n+80}{n+1}$$

$$\Rightarrow A_7 = a + 7d$$

$$\Rightarrow A_7 = 8 + 7 \left(\frac{24}{n+1} \right)$$

$$\Rightarrow A_7 = 8 + \frac{168}{n+1}$$

$$\Rightarrow A_7 = \frac{8n+8+168}{n+1}$$

$$\Rightarrow A_7 = \frac{8n+176}{n+1}$$

Put A_7 & A_3 in (1)

$$\therefore \frac{8n+80}{n+1} = \frac{3}{5}$$

$$\Rightarrow \frac{8n+80}{8n+176} = \frac{3}{5}$$

$$\Rightarrow 5(8n+80) = 3(8n+176)$$

$$\Rightarrow 40n + 400 = 24n + 528$$

$$\Rightarrow 40n - 24n = 528 - 400$$

$$\Rightarrow 16n = 128$$

$$\Rightarrow n = 8$$

AP

(84)

- 4) Find the sum of all integers b/w 280 and 350 which are exactly divisible by 9.

Sol

Alc to given condition:

$$288 + 297 + 306 + \dots + 342$$

$$\therefore S_n = \frac{n}{2} \{a + l\}$$

For n :

$$\therefore a_n = a + (n-1)d$$

here,

$$a = 288, a_n = 342$$

$$d = 297 - 288 = 306 - 297 = 9$$

$$\therefore 342 = 288 + (n-1)9$$

$$\Rightarrow 54 = 9n - 9$$

$$\Rightarrow 63 = 9n$$

$$\Rightarrow n = 63/9 = 7$$

Now,

$$\Rightarrow S_n = \frac{1}{2} (288 + 342)$$

$$\Rightarrow S_n = \frac{1}{2} (630)$$

$$\Rightarrow S_n = 7(315)$$

$$\Rightarrow S_n = 2205$$

Ans

- 5) Find no. of terms if $a = 8$, $a_n = 1/512$ & $r = 1/2$.

Sol

$$\therefore a_n = ar^{n-1}$$

here,

$$a = 8, a_n = 1/512$$

$$r = 1/2 \text{ and } n = ?$$

$$\therefore 1/512 = (8)(1/2)^{n-1}$$

$$\Rightarrow 1/4096 = (1/2)^{n-1}$$

$$\Rightarrow (1/2)^{12} = (1/2)^{n-1}$$

$$\text{Thus, } 12 = n - 1$$

$$\Rightarrow n = 13$$

- 6) The yearly depreciation of certain machine is 20% of the value at the beginning of year. If the original cost of the machine is Rs. 50,000, find the value after 5 years.

Sol

(85)

$$\therefore a_n = ar^{n-1}$$

here,

$$a = 50,000, n = 6$$

$$r = 100\% - 20\% = 80\% = 0.8$$

$$a_6 = ?$$

$$\therefore a_6 = (50000)(0.8)^{6-1}$$

$$\Rightarrow a_6 = (50000)(0.8)^5$$

$$\Rightarrow a_6 = 16384$$

Ans

⑦ Find r , if $S_{10} = 244S_5$ in G.P.

Sol

$$\Rightarrow S_{10} = 244S_5$$

$$\therefore S_n = \frac{a(1-r^n)}{1-r}$$

$$\therefore \frac{a(1-r^{10})}{1-r} = 244 \left[\frac{a(1-r^5)}{1-r} \right]$$

$$\Rightarrow 1-r^{10} = 244(1-r^5)$$

$$\Rightarrow 1-(r^5)^2 = 244(1-r^5)$$

$$\Rightarrow (1-r^5)(1+r^5) = 244(1-r^5)$$

$$\Rightarrow 1+r^5 = 244$$

$$\Rightarrow r^5 = 243$$

$$\Rightarrow r = (243)^{1/5}$$

$$\Rightarrow r = 3$$

⑧ If A.M and H.M b/w 2 numbers are 5 and $4\frac{1}{5}$ respectively then find numbers.

Sol

$$\Rightarrow A.M = 5 \text{ --- (1)}$$

$$\Rightarrow H.M = 21/5 \text{ --- (2)}$$

let a & b be 2 nos.

Take (1)

$$\Rightarrow \therefore \frac{a+b}{2} = A.M$$

$$\therefore \frac{a+b}{2} = 5$$

$$\Rightarrow a+b = 10$$

$$\Rightarrow a = 10 - b \text{ --- (1)}$$

Take (2)

$$\therefore H.M = \frac{2ab}{a+b}$$

$$\therefore \frac{2ab}{a+b} = \frac{21}{5}$$

$$\therefore a+b = 10$$

(86)

$$\therefore \frac{2ab}{10} = \frac{21}{5}$$

$$\Rightarrow ab = 21$$

Put a in above eq.

$$\Rightarrow b(10-b) = 21$$

$$\Rightarrow 10b - b^2 = 21$$

$$\Rightarrow b^2 - 10b + 21 = 0$$

$$\Rightarrow b^2 - 7b - 3b + 21 = 0$$

$$\Rightarrow b(b-7) - 3(b-7) = 0$$

$$\Rightarrow (b-7)(b-3) = 0$$

either,

$$b=7 \text{ or } b=3$$

At $b=7$ we get,

$$\Rightarrow a = 10-7 = 3$$

At $b=3$ we get,

$$\Rightarrow a = 10-3 = 7$$

Thus, 3 and 7 are the req. 2 numbers.

9) If G.M and H.M b/w 2 nos are 15 and $5\frac{10}{13}$ respectively then find nos.

Sol

$$\Rightarrow G.M = 15 \text{ — (1)}$$

$$\Rightarrow H.M = \frac{75}{13} \text{ — (2)}$$

let a & b are 2 nos.

Take (1)

$$\therefore G = \pm \sqrt{ab}$$

$$\therefore \pm \sqrt{ab} = 15$$

$$\Rightarrow ab = 225 \text{ — (1)}$$

Take (2)

$$\therefore H.M = \frac{2ab}{a+b}$$

$$\therefore \frac{2ab}{a+b} = \frac{75}{13}$$

Put ab in above eq.

$$\Rightarrow \frac{2(225)}{a+b} = \frac{75}{13}$$

$$\Rightarrow \frac{2(225) \times 13}{75} = a+b$$

$$\Rightarrow 2(3) \times 13 = a+b$$

$$\Rightarrow a+b = 78$$

$$\Rightarrow a = 78-b \text{ — (2)}$$

Put value of a in (1)

$$\therefore (1) \Rightarrow b(78-b) = 225$$

$$\Rightarrow 78b - b^2 = 225$$

(87)

$$b^2 - 78b + 225 = 0$$

$$b^2 - 75b - 3b + 225 = 0$$

$$b(b-75) - 3(b-75) = 0$$

$$(b-75)(b-3) = 0$$

either,

$$b = 75 \text{ or } b = 3$$

Put b in (2)

At $b = 75$ we get,

$$\Rightarrow a = 78 - 75 = 3$$

At $b = 3$ we get,

$$\Rightarrow a = 78 - 3 = 75$$

Thus 3 and 75 are req. 2 numbers.

(10) The second term of H.P is $\frac{1}{2}$ and 5th term is $-\frac{1}{4}$ - Find 12th term.

Sol

In A.P:

$$\Rightarrow a_2 = 2 \text{ --- (1)}$$

$$\Rightarrow a_5 = -4 \text{ --- (2)}$$

Take (1)

$$\therefore a + (n-1)d = a_n$$

$$\therefore a + (2-1)d = 2$$

$$\Rightarrow a + d = 2$$

$$\Rightarrow a = 2 - d \text{ --- (3)}$$

Take (2)

$$\therefore a + (n-1)d = a_n$$

$$\therefore a + (5-1)d = -4$$

$$\Rightarrow a + 4d = -4$$

Put value of a

$$\Rightarrow 2 - d + 4d = -4$$

$$\Rightarrow 3d = -6$$

$$\Rightarrow d = -2$$

Put d in (3)

$$\therefore (3) \Rightarrow a = 2 - (-2)$$

$$\Rightarrow a = 4$$

Now,

$$\Rightarrow a_{12} = a + (12-1)d$$

$$\Rightarrow a_{12} = 4 + (11)(-2)$$

$$\Rightarrow a_{12} = 4 - 22 = -18$$

In H.P:

$$a_{12} = -\frac{1}{18}$$